Seat No.:_____

AHU ID No. ____________ Name ____________ Major ____________ Grade ____________

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Stony Brook Institute at Anhui University

2023 Fall Semester

Finite Mathematical Structures Final Exam

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	Total score
Score				
Grader				

I. Completions (5 items, 4 points each, 20 points total)

Score	

1. The number of ways that 5-letter sequences (formed from the 26 letters in the alphabet, with repetition allowed) contain exactly two A's and exactly one N is ____________.

2. Find the exponential generating function for a_r , the number of different arrangements of r objects chosen from 4 different types of objects with each type appearing at least two and at most five times.

   ____

3. How many ways are there to distribute 8 balls into 6 different boxes with the first two boxes collectively having four balls if the balls are distinct?

   ____

4. Combine the following terms using the identity formulas:

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2 = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_.$$

5. Find a recurrence relation for $a_{n,k}$, the number of ways to select n identical hats from k different boxes of hats with between one and four hats from each box ___________________________.

II. Calculations and Proofs (32 points)

Score	

6. (1 item, 8 points) Find the coefficient of x^{28} in $(x^3 + \ldots + x^8)^5$.

7. (3 items, 9 points) Consider the problem of counting the ways to distribute 31 identical objects in 6 different boxes with at least three objects in each box.

a) Model this problem as an integer-solution-of-equation-problem.

b) Model this problem as a certain coefficient of a generating function.

c) Solve this problem.

8. (3 items, 9 points) How many arrangements of 12 letters in ASSOCIATIONS have ALL of the following properties:

(a) there are at least 2 letters in between each S,

(b) vowels are in alphabetical order, and

(c) the arrangement starts with an A.

9.(1 item, 6 points) Find a recurrence relation for a_n, the number of sequences of 1's, 2's, and 4's summing to n with the subsequence 421 not allowed.

III. Comprehensive Analysis Problem (48 points)

Score	

10. (1 item, 8 points) How many ways are there to assign 8 different visiting students from Anhui University to one Stony Brook summer course, choosing from calculus I, linear algebra, or basket-weaving, if each course must have at least one student from Anhui University?

11. (2 items, 10 points)

a) Find a recurrence relation for a_n, the number of sequences of red flags, white flags and yellow flags along an n-foot high flagpole if red flags are 1 foot tall, white flags are 2 foot tall and yellow flags are 3 feet tall and no two red flags can be side-by-side.

b) Repeat part a) but now with the constraint that no red flag can be followed by a yellow flag.

12. (1 item, 6 points) What form does a particular solution for the linear nonhomogeneous recurrence relation

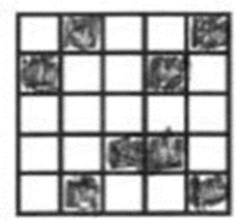
$$a_n = 4a_{n-1} - 4a_{n-2} + 2^n$$

13. (1 item, 8 points) Find the number of matchings of five men with five women given the constraints in the figure above on the right, where the rows represent the men and the columns represent the women. **You do not need to calculate the final number!**

(Formula: The number of ways to arrange n distinct objects when there are restricted positions is equal to

$n! - r_1(B)(n-1)! + r_2(B)(n-2)! + \cdot\cdot\cdot + (-1)^k r_k(B)(n-k)! + \cdots + (-1)^n r_n(B)0!$

where the r_k (B)s are the coefficients of the rook polynomial R (x, B) for the board B of forbidden positions)



14.(1item, 6points) what is the solution of the recurrence relation

$$a_n = a_{n-1} + 2a_{n-2}$$

with $a_0 = 2$ and $a_1 = 7$?

15. (1 item, 10 points) How many ways are there to arrange the 26 letters of the alphabet in a row such that none of the following words are formed by consecutive letters in the arrangement: INCH, LOST, or THIN? Be sure to define your Ai's.   (Hint: Watch out for common letters in the words.)